

# Umar Akram Mirad

uamirad00@gmail.com — +20 1023651704 — [linkedin.com/in/umarrakram](https://www.linkedin.com/in/umarrakram) — [github.com/umarrakram](https://github.com/umarrakram)

## Professional Summary

---

iOS Developer with hands-on experience building and shipping production apps in **Swift, SwiftUI, UIKit, and Combine**. Shipped 2 apps to the App Store including an IDE-class code editor for iPhone and iPad. Eager to grow as part of a team while contributing strong fundamentals in architecture, performance, and Apple platform best practices.

## Education

---

**B.Sc. in Computer Science & Information Technology** (Expected Jun 2026)

Egypt-Japan University of Science and Technology — Alexandria, Egypt

*Relevant Coursework:* Mobile App Development, Data Structures & Algorithms, Software Engineering, Machine Learning

## Core Skills

---

**Languages/Frameworks:** Swift, SwiftUI, Combine, UIKit, Python, C++

**iOS SDKs:** Core Data, Core Animation, Core Location, WebKit, PassKit, StoreKit, AVFoundation

**Architecture:** MVVM, Clean Architecture, Dependency Injection, Swift Concurrency (async/await, actors)

**Networking:** REST/JSON, URLSession, Vapor (server-side Swift), Codable

**Tooling:** Xcode, Instruments (Time Profiler, Allocations, Leaks), Firebase, Git, TestFlight, Postman

**Testing:** XCTest, XCUITest, snapshot tests

**Other:** Localization (Arabic/English), Accessibility, Agile/Scrum, App Store deployment

## Selected Projects

---

**Axios — IDE-Class Code/Text Editor for iPhone & iPad**     *SwiftUI, UIKit, Combine, WebKit, C++, Node.js*

[Link on App Store](#)

- Built and shipped a **complete text/code editor** to the App Store, designed as a practical **VS Code replacement on iOS** — sole developer from architecture to release.
- Implemented custom **text highlighting algorithms** (incremental lexing + on-the-fly tokenization) for responsive, large-file editing; optimized attributed-string pipelines and layout invalidation for 60 fps scrolling.
- Integrated **Node.js runtime** for project execution (**npm** workflows, GitHub repo cloning, package management) with sandboxed I/O; bridged output to SwiftUI terminals via Combine publishers.
- Embedded **C++ toolchain/compilers** for offline experiments; bridged to Swift with **extern "C"** shims and background execution via Swift Concurrency (async/await) and GCD.
- Added multi-pane editing (split view), search/replace with regex, minimap, auto-completion hooks, bracket matching, snippets, theming, and command palette.
- Crafted a robust file abstraction (local + iCloud Drive) with safe saves, conflict detection, and background indexing.
- Managed full release lifecycle: versioning, TestFlight distribution, App Store review, and post-launch crash triage via Instruments and analytics.

**WeatherSnap — Forecast App**

*SwiftUI, Combine, Core Location, OpenWeatherMap*

- Built a weather app with a reactive Combine pipeline (debounce, retry, backpressure) and GPS-based forecasts using **Core Location** with robust permission and error handling.
- Used **async/await** with **AsyncSequence** for background data refresh; implemented offline caching with stale-data policies and graceful degradation when network is unavailable.
- Designed composable SwiftUI components with Lottie-powered weather transitions; ensured accessibility support and Dynamic Type scaling.

## Professional Experience

---

### Onvo

iOS Developer

[Link on App Store](#)

*Remote - Egypt*

*Feb 2024 – Nov 2024*

- Designed and shipped SwiftUI-based social/creative features (profiles, music/movie suggestions, drawing messages), contributing to a **40% lift in DAU** alongside the product team.
- Implemented **Apple Pay (PassKit)** and **StoreKit** for one-time & subscription purchases; added receipt validation and server-side verification, reducing payment failures.
- Refactored network layer and REST integration with **Vapor** back-end: introduced response caching, request coalescing, and retry/backoff — measurably improving API response times.
- Built adaptive streaming via **WebKit** and AVFoundation; tuned buffer sizes to reduce video startup latency.
- Migrated legacy MVC modules to **MVVM + Combine**; introduced dependency injection and snapshot testing to improve code coverage and simplify rollbacks.
- Used Instruments (leak detection, time profiler) for performance profiling; triaged crashes and maintained performance budgets for cold start, scroll FPS, and memory.
- Participated in code reviews and knowledge-sharing sessions on Swift Concurrency and diffable data sources; aligned UI with Apple's HIG and accessibility best practices.

## Notable Achievements

---

- **2 iOS apps published** on the App Store with cumulative **100K+ downloads** and ~4.4/5 rating history.
- Sole developer of Axios, a shipped code editor used as a daily driver by iOS developers on iPad — demonstrates end-to-end ownership from architecture to App Store release.